

A disproof of Conjecture 00000000982

Fixed points of the Berezin transform on Fock space are neither radial nor at most two

1 The conjecture and its reading

Conjecture 00000000982 reads: *Definition: The Berezin transform. Conjecture: Fixed points of the Berezin transform on Fock space are radial functions, and the cardinality of the fixed-point set is at most 2 (fixed-point rigidity).*

It is a conjunction of two clauses: (A) every fixed point is radial, i.e. depends only on $|z|$; (B) the fixed-point set has cardinality at most 2. We use the classical Fock space of one complex variable. The phrase “on Fock space” leaves the class of admissible symbols open, so we treat two classes:

(R1) symbols in the Fock space F^2 itself;

(R2) bounded symbols, the standard domain of the Berezin transform of a function.

We show that (A) and (B) both fail under (R1), and (B) fails under (R2). Under (R2), clause (A) is true, because bounded fixed points are constant (see Section 4). The conjunction therefore still fails under (R2).

2 Definitions

Definition 1 (Fock space). Let dA be Lebesgue area measure on $\mathbb{C}$ and $d\lambda(w) = \pi^{-1}e^{-|w|^2} dA(w)$. The Fock space F^2 is the space of entire functions f with $\int_{\mathbb{C}} |f|^2 d\lambda < \infty$. Its reproducing kernel is $K(w, z) = e^{w\bar{z}}$, with $K(z, z) = e^{|z|^2}$. The normalized kernel is $k_z(w) = K(w, z)/\sqrt{K(z, z)} = e^{w\bar{z}-|z|^2/2}$.

This agrees with the Segal–Bargmann space as described on Wikipedia, where F^2 is “the space of holomorphic functions F in n complex variables satisfying the square-integrability condition” $\pi^{-n} \int |F(z)|^2 \exp(-|z|^2) dz < \infty$. There the kernel functions are $F_a(z) = \exp(\bar{a} \cdot z)$, with $\|F_a\|^2 = \exp(|a|^2)$, so $k_z = F_z/\|F_z\|$.

Definition 2 (Berezin transform). For a symbol $f : \mathbb{C} \rightarrow \mathbb{C}$, $(Bf)(z) = \int_{\mathbb{C}} f(w) |k_z(w)|^2 d\lambda(w)$. For bounded f this equals $\langle T_f k_z, k_z \rangle$, where T_f is the Toeplitz operator. This is the same pattern as the Bergman-disk formula $(Bf)(z) = \int_D \frac{(1-|z|^2)^2}{|1-z\bar{w}|^4} f(w) dA(w)$ quoted on Wikipedia.

A function f is *radial* if $|z| = |w|$ implies $f(z) = f(w)$. In Lean, the integral is Mathlib’s Bochner integral with respect to the Lebesgue measure `volume` on $\mathbb{C}$. A non-integrable integrand has integral 0 there. This convention does not create spurious fixed points in the classes we use. Integrability of $w \mapsto f(w)|k_z(w)|^2$ depends on z , and for the classes in the statement it holds at every z . For f in the Fock space, Cauchy–Schwarz bounds $\int |f||k_z|^2 d\lambda$ by $\|f\| \| |k_z|^2 \|$, which is finite. For a bounded symbol, $|f||k_z|^2 \leq \|f\|_{\infty} |k_z|^2$ and $|k_z|^2 d\lambda$ is a probability measure. A bounded fixed point

$f = Bf$ is automatically continuous, hence measurable, because Bf is a Gaussian convolution. In any case, the disproof only uses the constants and the monomials z^k , and for these every integral is shown to be integrable.

3 The proof

Lemma 3 (Gaussian convolution form). $|k_z(w)|^2 e^{-|w|^2} = e^{-|w-z|^2}$. Consequently $(Bf)(z) = \int_{\mathbb{C}} f(z+u) G(u) dA(u)$, where $G(u) = \pi^{-1} e^{-|u|^2}$.

Proof. We have $|k_z(w)|^2 = e^{2\operatorname{Re}(w\bar{z}) - |z|^2}$ and $|w-z|^2 = |w|^2 + |z|^2 - 2\operatorname{Re}(w\bar{z})$. Then substitute $w = z + u$; Lebesgue measure is translation invariant. $\square$

Lemma 4 (Gaussian moments). (i) $\int_{\mathbb{C}} G dA = 1$.

(ii) $u \mapsto |u|^k e^{-|u|^2}$ is integrable for every $k \geq 0$.

(iii) $\int_{\mathbb{C}} u^j G(u) dA(u) = 0$ for every $j \geq 1$.

Proof. (i) Mathlib's $\int_V e^{-b\|v\|^2} = (\pi/b)^{\dim V/2}$ with $V = \mathbb{C}$ gives $\int_{\mathbb{C}} e^{-|u|^2} dA = \pi$.

(ii) We have $x^k \leq k! e^x$ and $x - x^2 \leq \frac{1}{2} - \frac{1}{2}x^2$. Hence $|u|^k e^{-|u|^2} \leq k! e^{1/2} e^{-|u|^2/2}$, and the right side is integrable.

(iii) Let $\omega = e^{i\pi/j}$. Rotation $u \mapsto \omega u$ is a linear isometry of $\mathbb{C} = \mathbb{R}^2$, so it preserves Lebesgue measure, and $G(\omega u) = G(u)$. Hence $I = \int u^j G = \int (\omega u)^j G(\omega u) = \omega^j I = -I$, so $I = 0$. $\square$

Theorem 5. (a) B fixes every constant: $B(c) = c$.

(b) B fixes every monomial: $B(w^k) = w^k$ for all $k \geq 0$.

(c) Every constant and every monomial w^k lies in F^2 .

(d) $w \mapsto w$ is not radial.

(e) The monomials are linearly independent.

Proof. (a) Use Lemma 3 and Lemma 4(i).

(b) By the binomial theorem, $(z+u)^k = \sum_m \binom{k}{m} z^m u^{k-m}$. Each term is integrable against G by Lemma 4(ii). By Lemma 4(iii), every term with $k-m \geq 1$ integrates to 0. The term $m=k$ integrates to z^k .

(c) Constants and monomials are entire. Moreover $|w^k|^2 \pi^{-1} e^{-|w|^2} = \pi^{-1} |w|^{2k} e^{-|w|^2}$ is integrable by Lemma 4(ii).

(d) $|1| = |i|$, but $1 \neq i$.

(e) The map $p \mapsto (z \mapsto p(z))$ is injective on $\mathbb{C}[X]$, because a polynomial vanishing on $\mathbb{C}$ is 0. It sends the monomial basis to the functions $w \mapsto w^k$. $\square$

Theorem 6 (Main theorem). *The conjecture is false.*

- Under (R1), $w \mapsto w$ is a non-radial fixed point in F^2 , so (A) fails. The fixed-point set contains every constant, so it is infinite and (B) fails. It also contains infinitely many linearly independent functions w^k , so (B) fails even if “cardinality” is read as “dimension”.
- Under (R2), the fixed-point set contains every constant, so it is infinite and (B) fails.

4 Literature check

The result is consistent with the literature. Casseli (arXiv:1903.05364) states: “We show that an L^p function, $p \in [1, +\infty]$, with respect to the Lebesgue measure is invariant under this transformation if and only if it is harmonic. From this we deduce that the only bounded fixed points of the Berezin transform of polyanalytic Fock spaces are constant functions.” The same paper notes: “For the classical Fock space, a Lipschitz estimate for the Berezin transform of a bounded function, together with a semigroup property, shows that a bounded fixed point of the Berezin transform must be constant”. Asghari, Cuckovic and Sahutoglu (arXiv:2508.13115) state: “It is known that the Berezin transform is well-defined on the polynomials in z and $\bar{z}$.” On the polynomial domain, $w \mapsto w$ is again a non-radial fixed point. The proof above does not depend on these papers.

5 Relation to the earlier submission

An earlier submission by earthking11 was merged and then removed in the re-audit (commit 541cf4fb). The re-audit’s reason was: “no Berezin/Fock objects; load-bearing theorem prose-only”. Its Lean used only `import Std`. It modelled z on the integer lattice and proved only that $(1, 0) \neq (0, 1)$ and that $1, z, z^2$ are distinct. The identity $B = e^{\Delta/4}$ and the fixed-point property appeared only in prose. The present submission defines in Lean the Fock space, its normalized kernel and the Berezin transform as an actual Lebesgue integral over $\mathbb{C}$. It proves the Gaussian convolution form, $B(c) = c$ and $B(w^k) = w^k$ from Mathlib’s Gaussian integral, as well as F^2 membership, non-radiality and the infinitude of the fixed-point set.

6 Lean formalization

The file is `lean/Conjecture982/Basic.lean` (namespace `C982`), and it imports only Mathlib.

- Definitions: `gaussDensity`, `InFock`, `fockKernel`, `normKernel`, `berezin`, `IsRadial`, `fockFixed`, `boundedFixed`.
- Lemmas:
 - `normKernel_eq` states $k_z = K(\cdot, z)e^{-|z|^2/2}$;
 - `kernel_weight` and `berezin_eq_conv` give Lemma 3;
 - `integral_G`, `integrable_moment` and `moment_zero` give Lemma 4;
 - `berezin_const`, `berezin_pow`, `const_inFock`, `pow_inFock`, `id_not_radial`, `monomials_linearIndependent` and `fixed_infinite` give Theorem 5 and the infinitude of the fixed-point set.
- Main theorem `conjecture_982_false`. It states that $(w \mapsto w) \in \text{fockFixed}$ and is not radial, and that every $w^k \in \text{fockFixed}$. It also states that the monomials are linearly independent, that `fockFixed` and `boundedFixed` are infinite, and that $\neg((\forall f \in \text{fockFixed}, \text{radial } f) \wedge |\text{fockFixed}| \leq 2)$ and $\neg(|\text{boundedFixed}| \leq 2)$, with cardinalities measured by `Set.encard`.
- Axiom audit: `#print axioms C982.conjecture_982_false` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.